

Guilherme Ruiz Ferreira, PhD

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Skills

Languages	Native Portuguese — Fluent English — Basic Spanish.
Data & Tools	Linux, Docker, Git, SQL, InfluxDB, Microsoft Office, COMSOL
Programming	Python, C/C++, Java, Swift, ROOT, Mathematica, $\text{\LaTeX}$, HTML/CSS, VBA
Electronics	Arduino, microcontrollers, nuclear instrumentation modules, data acquisition systems
Academic	Data analysis, research, scientific writing, teaching/tutoring
Aviation	Private Pilot License (PPL), ~ 100 flight hours

Work History

- 2021 – 2025**  **PhD in Particle Physics**, University of Manchester — UK
Developed and tested detector instrumentation, carried out experimental measurements and simulations, built and maintained analysis pipelines, conducted sensitivity studies, and presented and published results with international collaborators.
- 2020 – 2021**  **Network Control Manager**, Amazon Logistics — London, UK
Modelled warehouse operations, analysed data to resolve connectivity issues, automated workflows and reporting (Python, SQL, VBA), built internal web tools (HTML/CSS), and delivered actionable insights to stakeholders in a fast-paced environment.
- 2014 – 2016**  **Undergraduate Researcher**, University of São Paulo (USP) — Brazil
Contributed to detector construction and testing for a major collider experiment upgrade, participated in collaborative data taking and analysis, and presented research at an international symposium.

Education

- 2018 – 2021**  **MSc in Physics** — Grade: Distinction
University College London (UCL) — UK
- 2013 – 2018**  **BSc in Physics with Astronomy** — Grade: 7/10
University of São Paulo (USP) — Brazil

Research Publications

- 1 'Cosmic Ray Measurements Using Charge and Light Readout in a Pixelated Liquid Argon Time Projection Chamber'.  DOI: 10.48550/arXiv.2512.10830.
- 2 'Opportunities and challenges to study solar neutrinos with a Q-Pix pixel readout'.  DOI: 10.1103/xq7l-lv7j.
- 3 'First demonstration of a combined light and charge pixel readout on the anode plane of a LArTPC'.  DOI: 10.1088/1748-0221/19/11/p11010.
- 4 'Distinguishing circumstellar from stellar photometric variability in Eta Carinae'.  DOI: 10.1093/mnras/stz067.